Example 5.20

Evaluate: $\lim_{x\to 0}\frac{\sqrt{2+x}-\sqrt{2}}{x}$

Solution

$$\lim_{x\to 0}\frac{\sqrt{2+x}-\sqrt{2}}{x}$$

$$=\lim_{x\to 0}\frac{(\sqrt{2+x}-\sqrt{2})(\sqrt{2+x}+\sqrt{2})}{x(\sqrt{2+x}+\sqrt{2})}$$

$$=\lim_{x\to 0}\frac{2+x-2}{x(\sqrt{2+x}+\sqrt{2})}$$

$$=\lim_{x\to 0}\frac{1}{(\sqrt{2+x}+\sqrt{2})}$$

$$=\frac{1}{\sqrt{2}+\sqrt{2}}=\frac{1}{2\sqrt{2}}$$

Example 5.21

Evaluate: $\lim_{x\to a}\frac{x^{\frac{3}{5}}-a^{\frac{3}{5}}}{x^{\frac{1}{5}}-a^{\frac{1}{5}}}$

Solution

$$\lim_{x\to a}\frac{x^{\frac{3}{5}}-a^{\frac{3}{5}}}{x^{\frac{1}{5}}-a^{\frac{1}{5}}}=\lim_{x\to a}\frac{\frac{x^{\frac{3}{5}}-a^{\frac{3}{5}}}{x-a}}{\frac{x^{\frac{1}{5}}-a^{\frac{1}{5}}}{x-a}}$$

$$=\frac{\frac{3}{5}(a)^{-\frac{2}{5}}}{\frac{1}{5}(a)^{-\frac{4}{5}}}=3a^{\frac{-2}{5}+\frac{4}{5}}=3a^{\frac{2}{5}}$$

Example 5.22

Evaluate: $\lim_{x\to 0}\frac{\sin 3x}{\sin 5x}$

Solution

$$\lim_{x\to 0}\frac{\sin 3x}{\sin 5x}=\lim_{x\to 0}\left\{\frac{3x\times\frac{\sin 3x}{3x}}{5x\times\frac{\sin 5x}{5x}}\right\}$$

$$=\frac{3}{5}\frac{\lim_{x\to 0}\frac{\sin 3x}{3x}}{\lim_{x\to 0}\frac{\sin 5x}{5x}}$$

$$=\frac{3}{5}\left(\frac{1}{1}\right)=\frac{3}{5}$$

Example 5.23

Evaluate: $\lim_{x\to\infty}\frac{6-5x^2}{4x+15x^2}$

Solution

$$\lim_{x\to\infty}\frac{6-5x^2}{4x+15x^2}=\lim_{x\to\infty}\frac{\frac{6}{x^2}-5}{\frac{4}{x}+15}$$

$$=\frac{0-5}{0+15}=-\frac{1}{3}$$

Example 5.24

Evaluate: $\lim_{x\to\infty}x\tan\left(\frac{1}{x}\right)$

Solution

Let $y=\frac{1}{x}$ and $y\to 0$ as $x\to\infty$

$$\lim_{x\to\infty}x\tan\left(\frac{1}{x}\right)=\lim_{y\to 0}\frac{\tan y}{y}=1$$

Example 5.25

Evaluate: $\lim_{n\to\infty}\frac{\sum n^2}{n^3}$

Solution

$$\lim_{n\to\infty}\frac{\sum n^2}{n^3}=\lim_{n\to\infty}\frac{n(n+1)(2n+1)}{6n^3}$$

$$=\lim_{n\to\infty}\frac{1}{6}\left\{\left(\frac{n}{n}\right)\left(\frac{n+1}{n}\right)\left(\frac{2n+1}{n}\right)\right\}$$

$$=\frac{1}{6}\lim_{n\to\infty}\left\{\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)\right\}$$

$$=\frac{1}{6}(1)(2)=\frac{1}{3}$$

Example 5.26

Show that $\lim_{x\to 0}\frac{\log(1+x^3)}{\sin^3 x}=1$

Solution

$$\lim_{x\to 0}\frac{\log(1+x^3)}{\sin^3 x}$$

05_11th_BM-STAT_Ch-5-EM.indd 109 21-04-2020 12:12:41 PM